2023 Homework 2
Complete solutions and executable verification

EE3032 Introduction to VLSI Design

13 October 2026

The original two-page assignment is preserved in the assignments directory. The three questions are analytical. The accompanying Python and SPICE files provide additional reproducible verification without replacing the derivations.

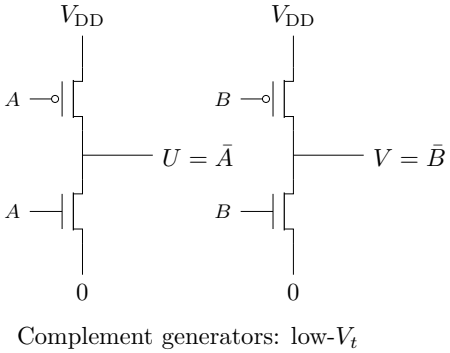
1. Multi-threshold CMOS design

The upper gate in Fig. 1 is XOR, preceded on its A input by an inverter. Consequently

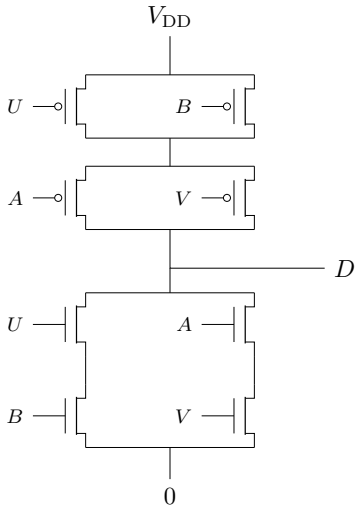
$$U = \bar{A}, \quad V = \bar{B}, \qquad \boxed{D = U \oplus B = AB + \bar{A}\bar{B}}, \qquad \boxed{E = \overline{BC}}.$$

With the stated one-unit low-V_t gate delay, the original paths are $A \to \text{INV} \to \text{XOR} \to D$ (two units), $B \to \text{XOR} \to D$ (one unit), and $B, C \to \text{NAND} \to E$ (one unit). The critical delay is two units. Use low-V_t devices in the A inverter and all devices realizing D. Use high-V_t devices in the four-transistor NAND for E, provided its characterized delay is at most two units. This spends the NAND's one unit of slack without extending the circuit's maximum delay.

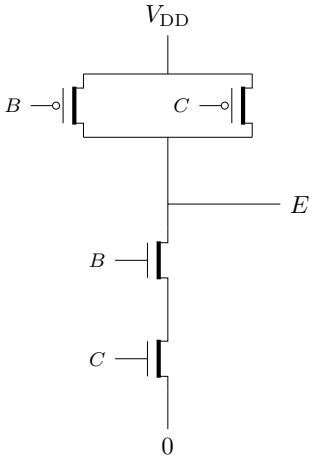
The following complete static CMOS implementation generates both complements locally. Its XOR core uses the pull-down condition $UB + AV$; the pull-up is the exact series/parallel dual. All shown thin channels are low-V_t; the four bold NAND channels are high-V_t. All pMOS bulks connect to V_{DD}, and all nMOS bulks to ground. No body-terminal connections are implicit internal-node ties.



Complement generators: low-V_t



D: low-V_t core



E: high-V_t NAND

The added B complement generator is kept low-V_t: in this specific static implementation the $B \to V \to D$ path also takes two gates. Merely selecting all B-controlled XOR devices as high-V_t would be

unjustified because those same series devices can carry a sensitized critical A transition. The prompt gives no high-V_t delay or leakage parameters, so an unconditional numerical timing guarantee or leakage reduction cannot be calculated. If the high-V_t NAND exceeds two units, retain low-V_t devices or resize and characterize it. The standard CIC018 TT model is not a dual-V_t model. `reproduce.py -logic-only` checks all eight (A, B, C) assignments.

2. Non-ideal effect

(a) Precharge

The on PMOS raises BL to V_{DD}. The access NMOS gate is at V_{DD}; charging the storage capacitance stops in the stated cutoff approximation when $V_{GS} = V_{\mathrm{DD}} - V_C = V_t$. Thus

$$\boxed{V_C(0) = V_{\mathrm{DD}} - V_t.}$$

The PMOS does not lose one NMOS threshold. If body effect is included, V_t is the effective threshold at the final source voltage, so the equation is implicit rather than a subtraction of the zero-body-bias threshold.

(b) Retention

With the switches off and the storage junction leaking a constant $I_L > 0$,

$$C\frac{dV_C}{dt} = -I_L, \qquad V_C(t) = V_{\mathrm{DD}} - V_t - \frac{I_L}{C}t.$$

Set this equal to the buffer threshold $V_{\mathrm{DD}}/2$:

$$\boxed{t_{\mathrm{ret}} = \frac{C(V_{\mathrm{DD}}/2 - V_t)}{I_L}, \quad V_t < V_{\mathrm{DD}}/2.}$$

If $V_t \geq V_{\mathrm{DD}}/2$, there is no initially valid stored high with positive margin. For zero leakage and positive initial margin this ideal model has infinite retention. The result includes the capacitance C, necessary dimensionally; the Lin answer omits it. The Liao and Jian answers include it.

In the printed topology the buffer observes BL, on the other side of the off access NMOS. Therefore this is the intended *stored-charge* retention, assuming an ideal nondestructive read. The continuously observed Out while both switches remain off cannot be inferred from C alone. BL capacitance, leakage and the read protocol are missing. Upon reconnecting an initially charged BL, even an ideal switch would give charge sharing $V_{\mathrm{share}} = (CV_C + C_{BL}V_{BL,0})/(C + C_{BL})$.

For a clearly labelled example, take $V_{\mathrm{DD}} = 1.8$ V, $V_t = 0.45$ V, $C = 20$ fF, $I_L = 1$ pA: initial voltage 1.35 V, switching boundary 0.90 V, and retention 9.00 ms. These values are illustrative, not specified assignment data. `q2_retention.sp` implements the ideal capacitor and current source; it is separate from the process transistor model.

3. Pseudo-NMOS NOR gate

(a) Function

The PMOS gate is grounded and its pull-up is always on. Two parallel NMOS devices pull down when either A or B is high. Therefore

$$\boxed{V_o : \ \overline{A + B}.}$$

This is ratioed logic with nonzero static current when the output is low. For an arbitrary reference aspect ratio S_0, let $s_p = (W/L)_p/S_0$, $s_A = (W/L)_{nA}/S_0$ and $s_B = (W/L)_{nB}/S_0$. The stipulated resistance model gives

$$R_P = 2R_n/s_p, \quad R_A = R_n/s_A, \quad R_B = R_n/s_B.$$

A	B	Pull-down	V_o/V_{DD}
0	0	open	1
1	0	R_A	$s_p/(s_p + 2s_A)$
0	1	R_B	$s_p/(s_p + 2s_B)$
1	1	$R_A \parallel R_B$	$s_p/[s_p + 2(s_A + s_B)]$

(b) The wording changes the answer

Unlike 2020 and 2021, the original 2023 PDF says *smallest* logic-zero voltage is $0.1V_{\mathrm{DD}}$. This is not silently replaced by "highest." For symmetric $r = s_A/s_p = s_B/s_p$, the smallest low is the two-on case:

$$V_{11}/V_{\mathrm{DD}} = \frac{1}{1+4r}.$$

If "smallest is" means equality, the literal answer is

$$\boxed{r = \frac{9}{4} = 2.25}, \qquad V_{01} = V_{10} = \frac{2}{11}V_{\mathrm{DD}} \simeq 0.1818V_{\mathrm{DD}}, \quad V_{11} = 0.1V_{\mathrm{DD}}.$$

These remain below a $0.5V_{\mathrm{DD}}$ decision boundary, but a receiver threshold is not specified in Question 3. If "smallest" means a lower bound $V_{11} \geq 0.1V_{\mathrm{DD}}$, it gives $r \leq 9/4$ and does not uniquely determine size. For unequal NMOS sizes the equality gives $s_A + s_B = (9/2)s_p$.

If the intended requirement instead matches the earlier years' conventional *maximum acceptable* low $V_{OL,\max} = 0.1V_{\mathrm{DD}}$, each one-on case must satisfy $s_p/(s_p + 2s_A) \leq 0.1$ and its B counterpart. Then

$$\boxed{(W/L)_{nA}, (W/L)_{nB} \geq \frac{9}{2}(W/L)_p.}$$

At the symmetric boundary $r = 4.5$, $V_{01} = V_{10} = 0.1V_{\mathrm{DD}}$ and $V_{11} = V_{\mathrm{DD}}/19$. These are ratios; absolute widths and lengths are free. Lin and Liao use this maximum-low interpretation. Jian discusses a lower bound for the two-on state, but its inequalities do not identify the same conventional sizing condition. Both interpretations are explicitly separated here instead of merging incompatible conditions.

Executable checks and reference comparison

`sizing.csv` records the two literal/maximum boundary ratios and an additional margin example. `q3_resistance.sp` measures all four divider cases at the conventional boundary ($R_P = 9$ kΩ, each NMOS resistance 1 kΩ). Its predicted voltages at 1.8 V are $(1.8, 0.18, 0.18, 1.8/19)$ V. `q3_cic018.sp` separately sweeps the actual BSIM3 devices at ratios 1, 2, 2.25, 3, 4.5, 4, 6 and 8. The order of the sweep is immaterial. BSIM3 does not enforce constant $R_P = 2R_n$; its voltages must not substitute for the stipulated divider answer.

The three supplied 2023 HW2 submissions (Lin 110501558, Liao 110501512, Jian 109501570) were read before derivation and compared above. Their threshold-retention model and NOR truth table are retained where valid. The complete transistor drawing, explicit dual-V_t timing condition, correct dimensional retention formula, BL qualification and the separate literal sizing interpretation improve the combined answer.

Run `python reproduce.py -work PATH` to regenerate the logic checks, three real HSPICE measurement tables and `validation.json`. Use `-model PATH` and `-hspice PATH` for other installations. The process file remains local. All stored measurements are actual HSPICE outputs; analytical checks are labelled separately.

Recorded execution

All three decks completed with PrimeSim HSPICE S-2021.09, with exit code zero. The eight logic cases passed. Ideal retention measured 9.000025 ms versus 9.000000 ms analytically; the 25 ns difference is startup integration, about 2.78×10^{-6} relative error. The four resistor results match the predicted values to the saved precision. The separate TT BSIM3 sweep uses $W_p = 0.5\,\mu$m, $L_n = L_p = 0.18\,\mu$m and $W_n = rW_p$:

r	$V_{01} = V_{10}$ (V)	V_{11} (V)	V_{00} (V)
1	0.176482	0.078389	1.800000
2	0.078122	0.036908	1.800000
2.25	0.068486	0.032568	1.800000
3	0.049900	0.024040	1.800000
4.5	0.032241	0.015732	1.800000
4	0.036566	0.017786	1.800000
6	0.023779	0.011676	1.800000
8	0.017604	0.008684	1.800000

These voltages quantify the chosen process-model example, not the exact constant-resistance size boundary. The transistor deck also saves supply currents and static power. The raw listings remain local; the compact CSV tables and reproduction scripts are included in the repository.